MIHIR MITHANI

VLSI Physical Design Intern | ASIC Flow · RTL Design · Verification

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SUMMARY

B.Tech ICT student (graduating Mar 2027) and current VLSI Physical Design Intern running the OpenROAD RTL-to-GDSII flow with TCL/shell automation. Built an ORFS desktop automation tool (ASAP7, Sky130, Nangate45, GF12), a self-checking SystemVerilog FIFO testbench (0 mismatches over 245 reads), and a cuOpt-based chip routing optimizer. Seeking fresher roles in physical design, RTL design/verification, and GPU-accelerated EDA.

EDUCATION

Marwadi University, Rajkot — B.Tech, Information and Communication Technology *Jul 2023 – Mar 2027*

CGPA: 8.88/10 | Coursework: Digital Design, CMOS Circuits, Computer Architecture, Computer Arithmetic, VLSI, Embedded Systems, ML/DL

Indian Public School, Surendranagar — 12th Science (PCB)

Jun 2021 – May 2023

EXPERIENCE

VLSI Physical Design Intern — Microcircuits Innovations Pvt. Ltd., Ahmedabad *Jun 2026 – Present*

- Execute the OpenROAD RTL-to-GDSII flow (floorplanning through routing) in a Linux/EDA tool environment.
- Automate EDA workflow steps with TCL and shell scripts.

Hardware Intern — eInfochips (An Arrow Company), Ahmedabad *Jun 2025 – Aug 2025*

- Designed and validated an IoT-based hardware prototype for restaurant operations, integrating sensor modules and relay-based control logic.
- Improved existing operational and debugging workflows, increasing system performance by over 30%.

TECHNICAL SKILLS

Physical Design: OpenROAD / OpenROAD Flow Scripts (ORFS) RTL-to-GDSII: floorplanning, placement, CTS, routing; SDC, STA fundamentals, KLayout; PDKs: ASAP7, Sky130, Nangate45, GF12
RTL & Verification: Verilog, SystemVerilog, self-checking testbenches, randomized testing, scoreboarding, Computer Architecture, Computer Arithmetic
Languages & Scripting: Python, TCL, Bash/Shell, C/C++
GPU / HPC: CUDA (Numba), NVIDIA cuOpt
Embedded & Robotics: NVIDIA Jetson Nano, ROS, NVIDIA Isaac Sim / Isaac Lab, TurtleBot, SO-100, IoT sensor and relay-based prototyping
AI/ML: LightGBM, Machine Learning and Deep Learning fundamentals
Tools: Linux CLI, Git/GitHub, Vim/GVim, EasyEDA

PROJECTS

OpenROAD Flow GUI — RTL-to-GDSII Desktop Automation Tool | github.com/Mihir-Mithani/OpenRoad_GUI *May 2026 – Jul 2026*

- Built a Python/Tkinter desktop app automating OpenROAD Flow Scripts (ORFS), the open-source RTL-to-GDSII ASIC flow.
- Implemented a PDK-aware wizard that auto-generates design directories (Verilog, SDC) for ASAP7, Sky130, Nangate45 and GF12.

Chip Routing Optimizer using NVIDIA cuOpt | github.com/Mihir-Mithani/CuOpt_NVIDIA_NIM *May 2026 – Present*

- Developed a PCB-style chip routing optimizer using octilinear A* pathfinding, integrated with NVIDIA cuOpt's VRP API to minimize total wirelength across the chip layout.

Synchronous FIFO — RTL Design & Verification | github.com/Mihir-Mithani/Synchronous_FIFO *Aug 2026*

- Designed a parameterized synchronous FIFO in SystemVerilog with full/empty flags and overflow/underflow protection.
- Built a self-checking testbench with a queue-based reference model: directed + 500 randomized ops, 245 reads scoreboarded with 0 mismatches; confirmed checker sensitivity via deliberate bug injection.

CERTIFICATIONS

- Digital Design Using Verilog — NPTEL | Computer Architecture — NPTEL | Verilog HDL Basics — Intel
- Fundamentals of Accelerated Computing with CUDA Python — NVIDIA DLI | Getting Started with AI on Jetson Nano — NVIDIA DLI

AWARDS & LEADERSHIP

- **Project Coordination & Phase Lead, Robotics Team** — ICT, Marwadi University (Sep 2025 – May 2026): led a 15-member team delivering SO-100 Arm and TurtleBot projects using ROS and NVIDIA Isaac Sim / Isaac Lab; optimized ROS-based communication pipelines.
- Rank 1, CodeJam Sem 6 (GeeksforGeeks) | Winner, Byte Hunt IoT Trail — Circuitology Club | IEEE Student Member